

ROBERT BASHIR MUKANDA

AI & Back-end Engineer — Python · Go · LLM Systems

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PROFILE

Final-year Computer Science student with a Diploma in Information Technology (JKUAT) and CCNA certification, building AI and back-end systems in production: a RAG pipeline on Qdrant, LLM agents connected to live business tools, and a concurrent Go service running across multiple clients. Built an offline-first fintech PWA that a Kenyan savings cooperative uses to manage member funds. Seeking a software or AI/ML internship, with a keen interest in product design.

SELECTED WORK & PROJECTS

AI-Powered Internship Outreach Automation System (BASHIR-2.0) | Python, Ollama, Google APIs, SQLite

Solo developer · in production · [GitHub](#)

- Architected a six-agent pipeline that automates discovering IT companies, extracting contact emails, writing personalised outreach emails and managing follow-ups.
- **Scout Agent:** uses the Google Places API (New) with zone-based search across 7 Nairobi zones and 40+ industry-specific queries to discover companies.
- **Hunter Agent:** crawls company websites with BeautifulSoup and aiohttp to extract email addresses through several strategies.
- **Composer Agent:** writes personalised attachment-request emails with a local Ollama LLM (Qwen, Llama), using prompt engineering for a professional tone; personal data never leaves the student's machine.
- **Mailer Agent:** creates Gmail drafts with the PDF CV attached through the Gmail API, with 30–90 s send throttling and daily send limits.
- **Tracker and Follow-up Agents:** sync every company, email and status to Google Sheets, detect replies and bounces, and send a polite follow-up after four days (at most two per company).

Simba Women SACCO — Offline-First Financial Ledger (PWA) | Vanilla JavaScript, PWA, Service Workers

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- Built and shipped a web app that digitised the paper ledger of a 32-member Kenyan savings cooperative managing KSh 1M+ in member savings and KSh 800K+ in active loans.
- Modelled the co-op's real financial rules in code: 20% per-cycle interest with flat arrears carry-forward, automatic default flagging after two missed cycles, and two-tier year-end dividends, reconciled against eight months of handwritten ledgers.
- Engineered for low-end Android on unreliable networks: one HTML file under 100 KB with no dependencies, installable as a PWA with a network-first service worker for full offline use and automatic updates.
- Added automatic on-device saving, JSON backup and restore for moving phones, one-tap monthly PDF statements, and a natural-language input for recording transactions ("Mary repaid 5000").

Wallet App (Balances) | Next.js 16, React 19, Redis (Upstash)

Developer · 2026 · [GitHub](#)

- Built a password-protected wallet for tracking balances across bank cards and savings jars, with per-account transaction history and grand totals.
- Stored amounts in cents to avoid rounding errors; a password locks every page and API route, and changing it signs out every device.
- Designed an iOS-style interface (frosted glass, bottom sheets, spring animations) that installs to the home screen and opens full-screen like a native app.

Urban Suits — Online Store UX/UI Design | three.js, GSAP

Interface designer · 2026

- Designed a 3D storefront where the catalogue wraps around the visitor as a sphere of cards, so browsing feels like walking along a tailor's wall of cloth.
- Designed the full flow from browsing to booking a fitting, with fabric close-ups on hover, a flat-grid view for scanning, a mobile layout and a reduced-motion mode.

RAG Pipeline | Python, Qdrant

Developer

- Built a retrieval-augmented generation pipeline that serves grounded answers from a Qdrant vector store.

Concurrent Go Service | Go

Developer

- Built a concurrent Go service running across multiple client deployments.

LEADERSHIP & COMMUNITY INVOLVEMENT

- **Facilitator, Innovation Docket — UNESCO Club JKUAT:** facilitate student activities, event coordination and advocacy initiatives within the university community.
- **Millennium Fellow — Millennium Fellowship,** Class of 2026.
- **Project Lead, SOUTH Civic Platform** (in development): working with a two-person cross-functional engineering team to architect and build a central web portal that connects youth with county government programmes in Nairobi.

TECHNICAL SKILLS

- **Strongest:** Go (concurrency), Python, RAG and LLM agent systems, RESTful API design, vector search (Qdrant)
- **Also:** JavaScript / TypeScript, Next.js, React, Node.js, HTML, C++, Progressive Web Apps and Service Workers, PHP, SQL (SQLite, MySQL), Redis, Supabase, LangChain, LlamaIndex, OpenAI and Anthropic SDKs
- **Design and front end:** UX/UI design, three.js, GSAP
- **CS foundations:** Data Structures & Algorithms, Systems Design, Concurrency, TCP/IP Networking
- **Infra and tools:** Google Cloud (GCP), VPS provisioning and hardening, Tailscale, Docker, Linux, Bash, Git, n8n, Ollama, Claude Code, Antigravity, monitoring and observability

EDUCATION

B.Sc. Computer Science — JKUAT	Sep 2024 – Jul 2027 (expected)
Coursework: Data Structures & Algorithms, Databases, Operating Systems, Networks, Software Engineering	
Diploma in Information Technology — JKUAT	Sep 2022 – Jul 2024 (completed)

CERTIFICATIONS

- Cisco Certified Network Associate (CCNA) — routing, switching and secure deployment, Cisco, 2025
- AI Fluency: Frameworks & Foundations — Anthropic Academy, 2026
- Knowledge Management for Young Professionals — United Nations University, 2026